Stair-Climbing Robotic Dog for Oxygen Therapy Patients

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An autonomous quadruped that follows a patient on supplemental oxygen, detects a staircase, and climbs it carrying the tank — freeing both hands for the railing.

Computer Science & Engineering · USF College of Engineering · Senior Capstone Design — Team Robodog (Project #6)

MOTIVATION

Patients on long-term oxygen therapy must haul a portable concentrator everywhere — stairs are the highest fall-risk, highest-exertion obstacle in the home.

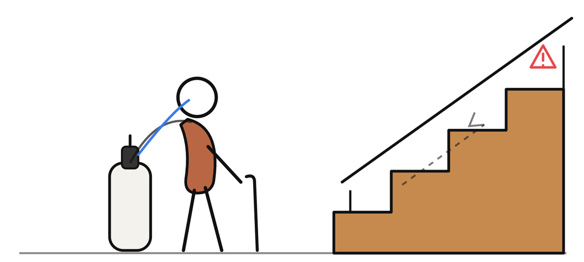

Fig. 1 — Patient must manage both tank and railing while short of breath.

- Older stair-users show higher stride variability & fall risk **(Startzell 2000; Hausdorff 1997)**.
- Our robot **carries the ~2.2 kg concentrator**, follows autonomously, detects stairs, and climbs — beside the patient.
- Both hands freed for the railing; robot handles the load.

HARDWARE PLATFORM

Robot: Unitree Go2, 6.9 kg, 12-DOF
Compute: NVIDIA Jetson Orin
Vision: Intel RealSense D435 RGB-D
LiDAR: Hesai XT16
Payload: 2.22 kg mock O_2 + cradle (~32 % of trunk mass)

Fig. 2 — Go2 with payload cradle.

SYSTEM ARCHITECTURE

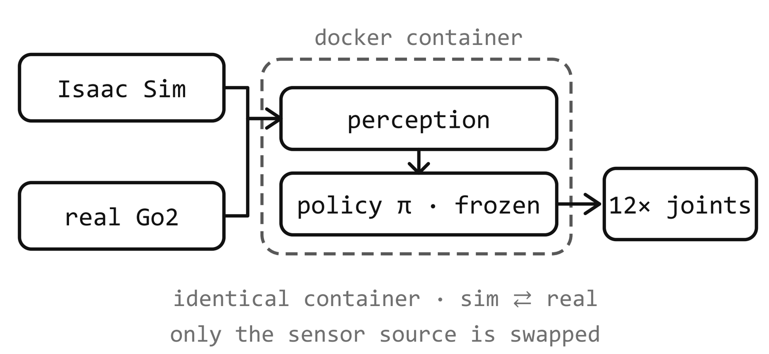

Fig. 3 — One shared codebase runs in NVIDIA Isaac Sim and on the real Go2; only the sensor source is swapped. Perception + control (Jetson / Docker) emit velocity commands; a shared policy library returns 12-DOF joint targets.

FROM PROVEN SIM TO THE LIVING ROOM

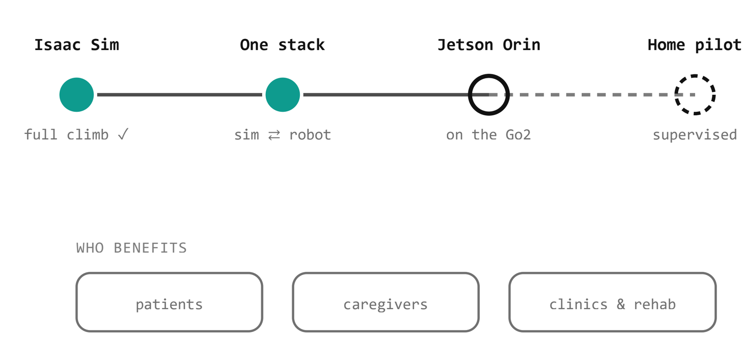

Fig. 4 — The same Docker control stack proven in Isaac Sim ports to the Jetson Orin on the real Go2; only the sensor source changes.

That is the deployment bet: a policy proven in simulation is expected to hold on hardware because it runs the identical container. Next up: hardware bring-up, gait polish, and a supervised home pilot.

METHODOLOGY: PERCEIVE → FOLLOW → DETECT → CLIMB

One control loop, running entirely on the robot's onboard Jetson Orin, turns raw camera and LiDAR frames into joint commands in five stages:

1. **Perceive** — RealSense D435 RGB-D + LiDAR feed two parallel vision heads: YOLOv8-Pose (person detection) and YOLO-World (open-vocab stair detection).
2. **Follow** — ByteTrack lock with 3 s occlusion hold. 3-zone PID holds 0.45 m standoff; depth-reactive avoidance routes around furniture.
3. **Detect Stairs** — YOLO-World counts risers; LiDAR depth corroborates (0.6 weight blend). ≥2 risers within 1.6 m triggers handoff. Dual-evidence gate eliminates false positives.
4. **Hand Off** — 50 Hz HandoffController (WALK ⇄ CLIMB → TOP-EGRESS / ABORT) swaps PGTT walker for blind-RL climb policy on motion-stall + riser trigger.
5. **Climb** — Proprioceptive "blind" RL policy drives legs up; at crest the robot walks clear and returns to walk policy.

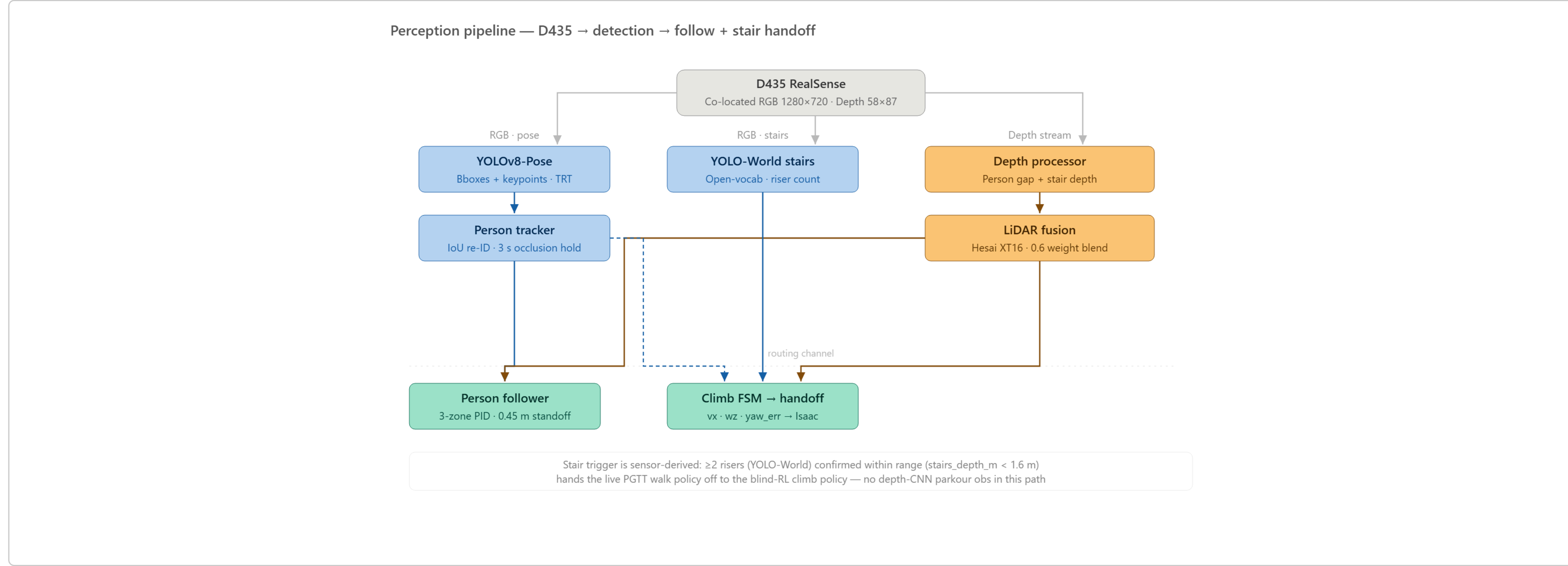

Fig. 5 — Perception pipeline: sensor inputs feed parallel vision heads; outputs merge in the Climb FSM.

RESULTS — ISAAC SIM STAIR-HEIGHT SWEEP

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Full climbs (0.12–0.15 m risers)

0.15 m
ADA-max riser successfully topped

≤31°
Peak body tilt (threshold: 60°)

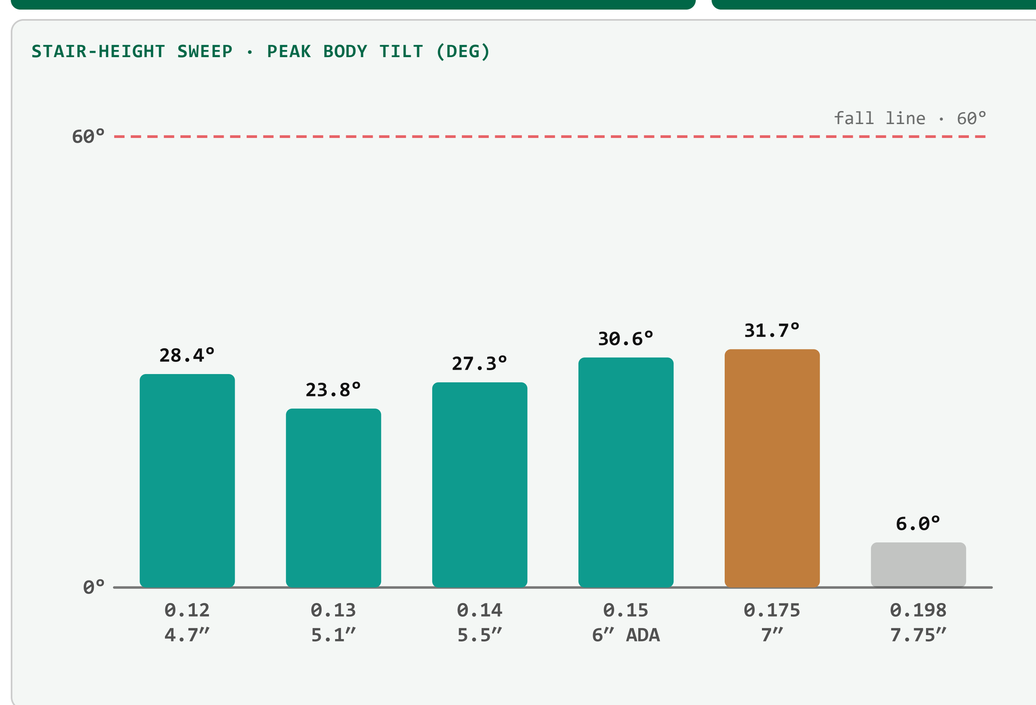

Fig. 6 — Peak body tilt at each riser height, well below the 60° fall line.

Fig. 7 — Climb time rises from 35.5 s to 51.2 s as the riser grows.

Riser	Steps (/ 14)	Peak Tilt	Outcome
0.120 m (~5″)	14.0	28°	✓ Full climb
0.130 m (~5″)	13.9	24°	✓ Full climb
0.140 m (~6″)	13.9	27°	✓ Full climb
0.150 m (6″ ADA)	14.0	31°	✓ Full climb
0.175 m (7″ IBC)	4.4	32°	✗ Partial
0.198 m (7.75″)	0.0	6°	✗ No climb

Robot topped the ADA-maximum (0.15 m) staircase end-to-end in simulation, staying upright throughout (peak tilt ≤31°, well below the 60° fall threshold).

WHY THIS MATTERS

About 1.5 million Americans rely on long-term home oxygen therapy, and stairs are consistently the task they fear most — a fall risk that can confine a patient to one floor of their own home. This project builds a sensor-guided quadruped that walks beside the patient, carries the oxygen equipment, and climbs the stairs on its own — so the patient never has to face a staircase alone.

CONTROL STATE MACHINES

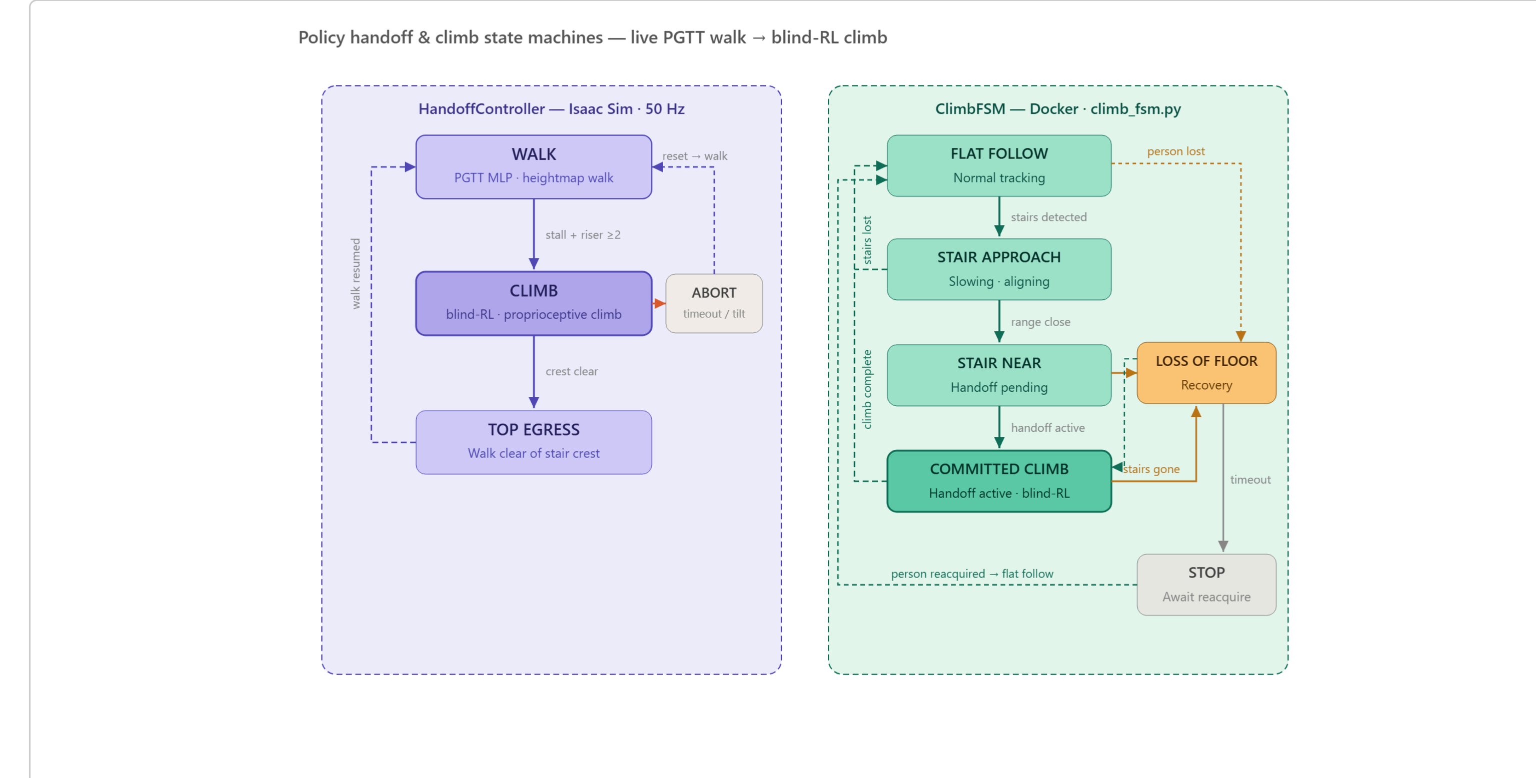

Fig. 8 — HandoffController (Isaac Sim, 50 Hz) and ClimbFSM (Docker) run concurrently via UDP.

SIMULATION & TRAINING

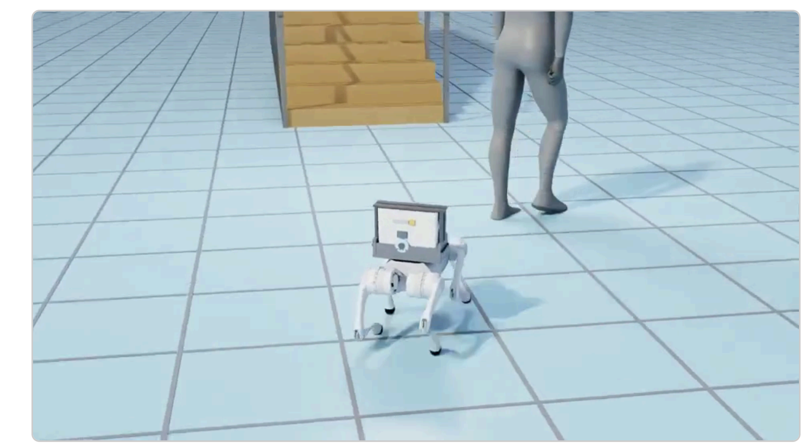

Fig. 9 — Scene view (Isaac Sim rollout): robot trails the patient toward the stairs, carrying the O_2 tank.

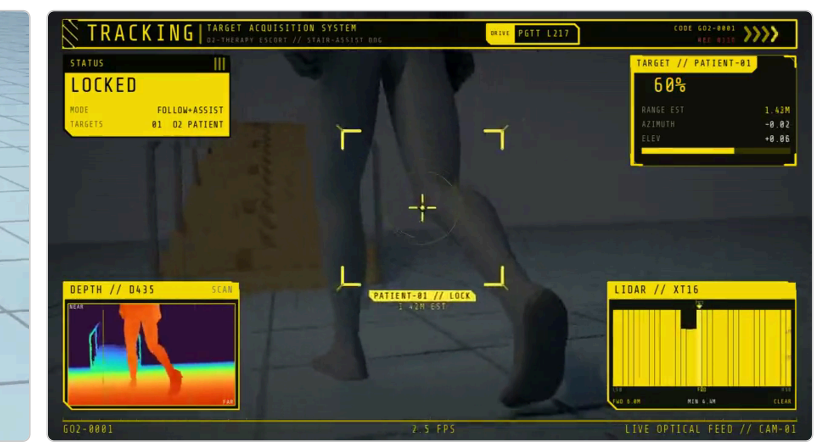

Fig. 10 — Live perception HUD: YOLO-World person lock, D435 depth, and LiDAR range panels.

BLIND-RL CLIMB POLICY

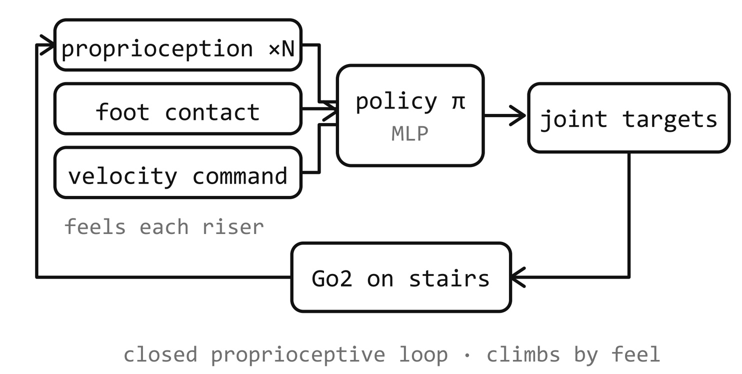

Fig. 11 — Blind-RL policy: no camera, no depth on the stairs — terrain is felt via proprioception + foot contact alone.

CONCLUSIONS

- End-to-end **follow** → **detect** → **hand-off** → **climb** stack validated in NVIDIA Isaac Sim on a shared sim/real codebase (**401 passing tests**).
- Successfully reaches the top of the **0.15 m (ADA-maximum)** staircase end-to-end in the best demo.
- Peak tilt ≤31° across all successful runs — **well below the 60° fall threshold**.

LIMITATIONS & FUTURE WORK

- **Nose-down gait** — blind-RL climber pitches forward; a gait-posture problem, not a handoff bug.
- **High CoG** — O_2 carrier raises center of gravity; wider stance being explored.
- **Safety grade-gate** — stance locking suppressed mid-climb; min observed clearance 0.209 m vs. 0.55 m target.
- **Hardware climb unproven** — physical Go2 validation is the active next step.

ACKNOWLEDGEMENTS

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